

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2023-24

Paper: CS-103 — Numerical Computing

Time: Three hours

Maximum Marks: 70

Instructions: Attempt ANY FIVE questions.

Question 1.

- (a) Diagonalize the given matrix:

$$\begin{bmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{bmatrix}$$

[7]

- (b) Solve the following equation using bisection method to find a real root between 2 and 3 (correct to 3 decimal places):

$$x \log_{10} x - 1.2 = 0$$

[7]

Question 2. For the following system of linear equations:

$$9x_1 + 2x_2 + 4x_3 = 20$$

$$x_1 + 10x_2 + 4x_3 = 6$$

$$2x_1 - 4x_2 + 10x_3 = -15$$

- (a) Check the convergence of the iterative procedure.

[4]

- (b) Use the Gauss-Seidel iterative procedure to solve the above (correct to 3 significant digits).

[10]

Question 3.

- (a) Derive the Lagrange interpolation formula. What are the advantages and disadvantages of using it?

[7]

- (b) Fit 3-degree Lagrange polynomial to find $f(x)$ at $x = 3.8$ for the following function:

x	1	2	3	4
$f(x)$	6	-3	6	2

Question 4.

- (a) Use the Newton's divided difference formula to derive:

(i) Trapezoidal Rule

(ii) Simpson's 1/3rd Rule

[7]

(b) Use Gauss-Legendre 2-point formula to calculate:

$$\int_0^1 (x^3 - 10x + 6) dx$$

[7]

Question 5. Solve the following differential equation (correct to 4 significant digits) using:

$$\frac{dy}{dx} = 2xy, \quad y(0) = 1, \quad 0 \leq x \leq 0.5, \quad h = 0.1$$

(a) Euler's Method

(b) Picard's Method

[14]

Question 6. For the given matrix:

$$\begin{bmatrix} 1 & 3 & 4 \\ 3 & 1 & 2 \\ 2 & 1 & 0 \end{bmatrix}$$

calculate:

(a) Inverse matrix using Gauss Elimination method.

[7]

(b) Largest Eigen value and corresponding Eigen vector using iterative method.

[7]

Question 7. Compute the root between 1 and 2 of the equation given below using:

$$x^3 - 2x^2 + 3x - 5 = 0$$

(a) Regula-Falsi method

[7]

(b) Newton Raphson method

[7]